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## <option> tag

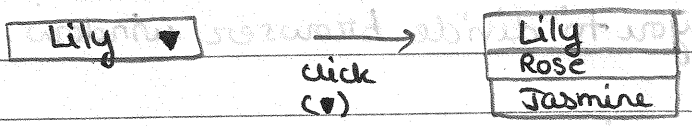
occurs only within select element or tag and is used to represent each choice of select.
it defines option that can be selected.

attribute → label (specifies shorter label for an option)

selected (defines default option to be selected when webpage is loaded)

value ( specifies value to be sent to server when form is submitted.

```
eg: <select name = "flowers">
    <option value = "lily"> Lily </option>
    <option value = "rose"> Rose </option>
    <option value = "jasmine"> Jasmine </option>
    </select>
```

output : 


## <textarea> tag

it is a container tag that defines multiline input text area/field. You need to specify rows and columns.

```
ex: <textarea name = "area" rows = "10" cols = "30">
    I was hanging out with friends.
    </textarea>
```

cols specifies width, rows specifies lines in text area.

output: column = 30

rows 10

I was hanging out with my friends.

<button> tag

defines clickable button.

eg: <button type = "button" onclick = "alert ('Hello Girls!')"> click </button>

output

click (button) → when pressed

Hello Girls! ok

alert message.

Frames

→ allows you to divide browser window into several parts.

→ way of displaying two or more pages at once.

<frameset> tag.

→ use this tag instead of <body> tag.

→ frame container for dividing window into rectangular subspaces called frames.

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attribute ⟶ cols (how many columns are contained in frameset)

↳ absolute values in pixels

< frameset cols = "100, 70" >

↳ percentage of browser

< frameset cols = "70%, 50%" >

↳ using wildcard symbol

< frameset cols = "70%, *" >

↳ takes remainder of window.

↳ relative width of browser window

< framset = "3*, 2*, 1*" >

: rows (used to specify rows)

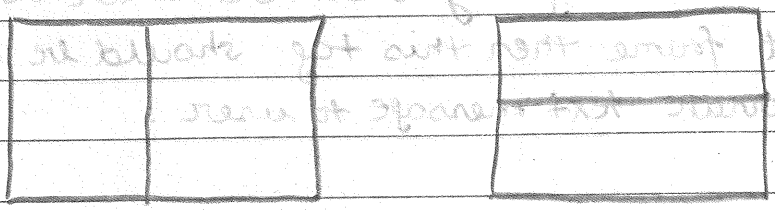
: border (specifies width of border of each frame in pixels)

eg: < frameset border = "5" >

: frameborder (specifies whether three dimensional border should be displayed b/w frames) < frameset frameborder = "0" >

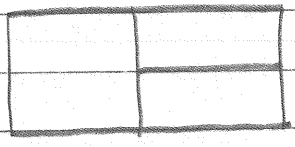
: framespacing (amount of space b/w frames in frameset)

< frameset framespacing = "8" >



browser window

dividing browser window in frames



<frame> tag

Each frame is indicated by <frame> tag and it defines which HTML document shall open into frame.

attribute. → src (used to give filename that should be displayed in frame)
Its value can be URL.

name (allows you to give name to frame)

marginwidth and marginheight: allows you to width and height of frame border and content. value in pixels.

noresize. (prevent a user from being able to resize frame)

scrolling: (controls appearence of scrollbars that appear on frame)
values → yes, no, auto.

<noframe>

if user is using any browser which doesnot support frame then this tag should be displayed for alternate text message to user.

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Example :

```
<html>
<head>
<title> frames </title>
</head>

<frameset cols = " 25%, 50%, 25% ">
<frame name = " Left "  src = " File1.html "/ >
<frame name = " center "  src = " File2.html " / >
<frame name = " Right "  src = " File3.html " / >

<noframe>
   <body> Your browser doesnot support frames
   </body>
</noframe>
</frameset>
</html>
```

Output :

| Left | center | Right |
|---|---|---|
| | | |

if <frameset rows = " 25%, 50%, 25% "> then,

Output :

| Left |
|---|
| Center |
| Right |

# Cascading Style sheet (CSS)



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CSS is a language that describes style of an HTML document
describes how HTML element should be displayed.
because HTML was created to describe content of webpage only. (<font> and color attributes were added to HTML, it started a nightmare for developers to add style to every single page became expensive).

Syntax: consist of selector and a declaration blocks:

declaration declaration

ex: h1 { font-size : 12px ; color : red ; }
selector property value property value

selector: points to HTML element you want to style.

declaration block contains one or more declarations seperated by semicolon.
Declaration includes CSS property name and a value, seperated by semicolon. (always end with semicolon)

CSS selectors → (element, class, id)

I) element selector
(based on element name)

eg: p {
direct by element name ↑ color : red;
text-align : center;
}

All <p> elements of page will be center aligned, red text color.

II) id selector (to select a specific element)

↳ should be unique within a page.

↳ write (#) character followed by id element.

eg:

```
#para1
{
    text-align: center;
    color: red;
}
```

will be applied to this only

should be unique.

```
<html> ....
<body>
<p id = "para1"> .... </p>
:
</body> </html>
```

III) class selector (select element with specific class attribute)

↳ use (.) dot character followed by name of class

```
.para1
{
    text-align: center;
    color: red;
}
```

applied to all para1

not unique ie can be applied to many

```
<html> ...
<p class = "para1"> .... </p>
<h1 class = "para1"> .... </h1>
:
</body>
</html>
```

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can also be applied like this

```
<html> ...
<body>
<h1 class = "center">...
</h1>
<p class = "center"> ...
</p>
:
</html>
```

```
p.center
{
  text-align : center;
  color : red;
}
```

will be applied only to <p> tag with class = "center".

You can also group selectors

```
ex:  h1, h2, p {
        text-align : center;
        color : red;
     }
```

CSS comments

Starts with /* */ ends with.

 tag

defines division or a section in an HTML document. This tag is used to group block-elements to format them with with CSS

```
eg: < div style = " color : red ">
      <h3> This is heading </h3>
      <p> This is para </p>
    </div>
```

output : display a section in a document in red color.

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Inserting CSS
there are three ways

I) External Stylesheet

- you can change look of an entire website by changing just one file.
- include a reference to external style sheet file inside <link> element.

eg:
```
<head>
  <link rel="stylesheet" type="text/css"
href="stylesheetname.css">
</head>
```

Stylesheet must be save with .css and should not contain any html tags.

II) Internal Stylesheet.

- used if one single page has a unique style.
- defined within <style> element, inside <head>.

eg:
```
<html>
<head>
<style>
 body {
        background-color : blue;
      }
 h1 {
      color : red;
    }
</style>
</head> .... <body>...</body> </html>
```

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III) Inline styles.

- used to apply a unique style for single element
- add style to relevant attribute.

eg:
```
<html>
  <body>
  <h1 style = " color : blue ; margin-left : 30px; ">
    ....
  </h1>
  <p> ... </p>
  </body>
  </html>
```

Cascading order (which style is used if more than one style is specified for element)

(i) Inline style. (highest priority)
(ii) External and internal stylesheets (in head section)
(iii) Browser default.

→ CSS color.

used to specify color either for foreground of an element or else for background of element.

(i) color : sets text color for different elements.

```
<p style = " color : red ; "> ... </p>
```

(ii) opacity : sets opacity level for  element

```
< div style = " opacity : 0.5 ; " > ... </div>
```

List of all possible format

| | Format | Syntax | Example |
|---|---|---|---|
| (i) | Hexcode | #RRGGBB | #FF0000 |
| (ii) | Short Hexcode | #RGB | #F00 |
| (iii) | RGB % | rgb(rrr%, ggg%, bbb%) | rgb(50%, 50%, 50%) |
| (iv) | RGB Absolute | rgb(rrr, ggg, bbb) | rgb(0, 255, 0) |
| (v) | keyword | aqua, black, white etc. | color : red; |

→ CSS Background (used to define background color/image etc ie. effects for elements)

(i) background-color (set background color)

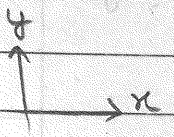
eg:
```
<body>
<p style = "background-color : red;" >
...
</p>
</body>
```

(ii) background image (set image at background)

eg:
```
body
{
  background-image : url;
}
```

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(iii) background-repeat (repeats an image both horizontally and vertically) → this property is used to control repetition of an image in background)




(i) repeat-x : repeat along x-axis
(ii) repeat-y : repeat along y-axis
(iii) repeat : both x and y axis
(iv) no-repeat : prevents from repeating at all.

(iv) background-attachment (sets background image is fixed or scrolls with rest of page).

```
eg: body
    {
      background-attachment : fixed ;
    }
```

→ CSS Fonts. (defines font family baldness, size, style of text)

(i) font-family (used to change face of font)

```
eg:
   <body>
   <p style = " font-family : serif ; " >
   ...
   </p>
   </body>
```

(ii) font-style (used to make a font italic or oblique.

```
eg: <p style = " font-style : italic ; " >
```

(iii) font-size (used to increase/decrease size of font)

eg: 
...

(iv) font-variant (used to create small caps effect)

eg: 
...

(v) font-weight (used to increase how bold or light a font appears)

eg: 
...

→ CSS Text (formatting text)

(i) word-spacing (used to add or subtract space b/w word of sentence)

eg:  ... 

(ii) letter-spacing (used to add or subtract space b/w alphabets of words)

eg:  ... 

(iii) text-decoration (used to underline, overline and strikethrough text)

eg:  ... 

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(iv) text-transform (used to capitalise text or convert to uppercase / lowercase)

eg:  ... 

(v) text-align (used to align text of a paragraph)

eg:  ... 

(vi) text-indent ( used to indent text of paragraph)

eg:  ... 

(vii) line-height ( used to specify space b/w lines)

eg:  ... 

→ CSS Border

allows you to specify border of box.

(i) Border-color (specifies color of border) — can take values (color-name, hex values (#ff0000), RGB values rgb(0,0,255), transparent)

(ii) Border-style ( specifies whether a border should be → solid, dotted, dashed, double, groove (3D), ridge (3D), inset (3D), outset (3D), none, hidden)

(iii) Border-width ( specifies width of border)
↳ can be in size (px, pt, cm, em etc) or values (thin, medium, thick).

eg:  ... 

→ CSS Margins (defines space around HTML element)
↳ it is possible to have negative value to overlap content

(i) Margin (specifies shorthand property for setting margin in one declaration)

(ii) Margin-bottom, Margin-top, Margin-left, Margin-right.
↳ for bottom ↳ for up ↳ for left ↳ for right

eg:  ... 

shorthand property (discussed in detailed)

(it is possible to specify all marjin property in one property.) can be applied to border, margin and padding

eg:  ... 
top right bottom left

 ... 
top right & left bottom

 ... 
top & bottom right & left

 ... 
↳ all has same value for all.

auto value : ( to horizontal center element within its container)

The element will take specified width, and remainig space will split equally b/w left & right margins.

 ... 

eg:  ... 
top right bottom left

(shorthand property for border) and so on as above.

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→ CSS Padding (used to add space around element's content, inside of any defined border)

Padding (shorthand), Padding-top (for top), padding-right (for right), padding-left (for left),

padding-bottom (for right)

Vertical-align : (align vertical an image)

eg: < img style = " Vertical-align : text-top; "> ... </img>

direction : (text direction to "right to left")

< div style = " direction : rtl; "> ... 

eg: < p style = " padding-top : 50px; padding-right : 50px; padding-bottom : 30px; padding-left : 20px; "> ... </p>

< p style = " padding : 25px 75px 50px 100px; ">
top right bottom left

Some Important points : values that can be taken

(1) Margin can take values

(a) auto - browser calculates margin

(b) length - px, pt, cm etc

(c) % - specifies margin in % of width

(d) inherit - specifies margin should be inherited from parent

Padding takes same value except "auto". padding cannot take "auto" value

→ CSS Layers

can create layers of various divisions

CSS layer refer to applying z-index property to element that overlap with each other.

The z-index property is used along with position property is used along with position to create an effect of layers. You can specify which element should come on top and which element should come at bottom.

<layer> tag can explicitly positioned by setting the LEFT & TOP attributes.

<ilayer> tag appear wherever they naturally fall in flow of document

→ Grouping stylesheets or CSS Combinations

A combinator is something that explains relationship b/w selectors. A CSS selector can contain more than one simple selector. B/w the simple selectors, we can include a combinator.

Different combinator are:

(i) Descendant Selector (space) : matches all elements that are descendants of specified element.

eg:
```
div p
{ background-color : red; }
```

(ii) child selector (>) : selects all element that are immediate children of specified element.

eg:
```
div > p
{ background-color : red;
}
```

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(iii) Adjacent sibling Selector (+) : selects all element that are adjacent sibling of specified element.

eg:
```
div + p
{   background-color : red;
}
```

(iv) Gereral sibling selector (~) : selects all element that are sibling of specified element.

eg:
```
div ~ p
{
    background-color : red;
}
```